

THE INTERN DIARIES

Intern Report

Data Quality Audit

WEEK 3 | DAILY JOURNAL ENTRY



LEARNING BY DOING, ONE DAY AT A TIME

Online Retail Dataset (RETAIL COMMERCE)

1. What were the major data-quality issues identified in the Online Retail II dataset?

Answer: The data-quality audit identified several issues in the Online Retail II dataset containing **1,067,371** records and 8 columns. The audit focused on missing values, duplicate records, range violations, date validity, and potentially inconsistent transaction values. This provided a structured view of the dataset before it was used for further analysis.

The most significant issue was missing Customer ID values, with **243,007** records lacking a customer identifier. This represents approximately **22.77%** of the dataset. This issue is particularly important for customer-level analysis because records without a customer identifier cannot reliably be associated with a specific customer for activities such as customer segmentation, retention, or cohort analysis.

The audit also identified **34,335** exact duplicate records, representing approximately **3.22%** of the dataset. In addition, there were **22,950** records with negative Quantity, 5 records with negative Price, and **6,202** records with a zero Price. These numerical anomalies were not all treated as definite errors because some may represent legitimate retail transactions such as returns, cancellations, promotional items, or adjustments.

Overall, the audit showed that the dataset is usable but requires different treatment for different quality issues rather than simply deleting every unusual record. The results demonstrate why data-quality analysis needs both automated validation rules and business interpretation.

2. Which data-quality issue required the greatest attention and why?

Answer: The issue requiring the greatest attention was the **missing Customer ID field**, with **243,007 missing records**, or approximately **22.77% of all rows**. This was considerably larger than the other detected issues and has a direct impact on any analysis that depends on identifying individual customers.

The reason this issue is particularly important is that Customer ID is not just another descriptive field. It is required to connect multiple transactions to the same customer. Without it, analyses such as customer retention, repeat purchasing, cohort analysis, and customer-level segmentation become incomplete because those records cannot be reliably attributed to a specific customer.

Compared with missing Customer IDs, the **34,335 duplicate records** affected a smaller proportion of the dataset, approximately **3.22%**. Negative Quantity affected **22,950 records**, or about **2.15%**, while negative Price occurred in only **5 records**. Therefore, the missing Customer ID issue has the largest numerical impact and also has important analytical consequences.

However, the issue was not handled by blindly deleting all 243,007 records. The appropriate response depends on the analysis being performed. For customer-level analysis, those records may need to be excluded, while for some transaction-level sales analysis they could still contain useful information. This distinction was an important part of the prioritization process.

3. What unusual transaction values were found, and were they treated as errors?

Answer: The audit identified **22,950 negative Quantity records**, **5 negative Price records**, and **6,202 zero-Price records**. These results show that the dataset contains several numerical values that require investigation before being treated as errors. A simple range check is useful for identifying them, but the meaning of each condition needs to be considered.

Negative Quantity values are particularly important in retail data because they can represent legitimate business events such as **returns or cancellations**. Therefore, the audit classified these records as a **review condition** rather than automatically marking all 22,950 records as corrupted data. This prevents the cleaning process from accidentally removing valid business transactions.

The **5 negative Price records** are much smaller in number and are more difficult to justify as ordinary positive-priced retail transactions. They were therefore treated as a structural quality problem for the cleaned analytical sample. The dataset also contained **6,202 zero-Price records**, which may represent free items, promotional transactions, adjustments, or other special cases, so these were retained for review rather than automatically deleted.

4. How much was the dataset improved after cleaning?

Answer: The cleaning process started with **1,067,371 rows**. Exact duplicate records were removed, and records with invalid dates or negative prices were excluded from the cleaned analytical dataset. After these clearly actionable cleaning rules were applied, the dataset contained **1,033,031 rows**, meaning **34,340 rows were removed**.

The majority of those removed records were exact duplicates, with **34,335 duplicate rows** identified during the audit. The remaining small difference came from the **5 records with negative Price**. No additional rows were removed because of invalid dates, since the dataset contained **0 invalid InvoiceDate values after datetime validation**.

Importantly, the cleaning process did **not** remove all missing Customer IDs, negative quantities, or zero-price transactions. These conditions were retained because they may have legitimate business meanings and were instead marked as issues requiring review. This preserves useful information while clearly documenting the quality concerns.

A **10,000-row reproducible cleaned sample** was then created for submission using a fixed random seed. The sample was **approximately 0.94 MB** and included a QualityFlag field to indicate whether records passed the checks or contained conditions such as missing Customer ID, negative Quantity, or zero Price. This makes the cleaned sample transparent and traceable.

5. What did the data-quality audit achieve overall?

Answer: The main achievement of the audit was the creation of a **repeatable quality-checking process** rather than a one-time manual inspection. Validation rules were applied to identify missing values, duplicates, numerical range issues, date validity, and other transaction-level conditions. Each issue was quantified by both count and percentage so that its impact could be evaluated objectively.

The audit established several clear quality results. There were **0 invalid InvoiceDate values**, confirming that the transaction dates could be parsed successfully. There were **34,335 exact duplicate rows, 243,007 missing Customer IDs, 22,950 negative Quantity records, 5 negative Price records, and 6,202 zero-Price records**. These figures provide an evidence-based picture of the dataset's quality.